

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO1: Analyze the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)

QUESTIONS: 20

Total Marks:100

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I A.Sai Rohit (192472144) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A.Sai Rohit

Annexure A

Short Answer Test

Answer to all :

20 X 5 = 100

1. A college wants to identify students who may fail in the final exam. How can Data Science help?
2. A hospital has patient records for many years. How can Data Science be applied?
3. An online shopping website wants to recommend products to customers. Which use of Data Science is applied here?
4. A bank wants to detect suspicious transactions. How can Data Science be useful?
5. A company wants to reduce customer loss. How can Data Science support this?
6. A school wants to improve student performance. Mention one benefit of Data Science in this case.
7. A supermarket wants to know which products sell more during festivals. How can Data Science help?
8. A transport department wants to reduce traffic congestion. How can Data Science be applied?
9. A farmer wants to improve crop yield. Give one use of Data Science.
10. A company has data in Excel tables, customer reviews, images, and videos. Which facet of data is shown here?
11. A weather app updates temperature every minute. Which facet of data is involved?
12. A university stores student details in rows and columns. What type of data structure is this?
13. Customer comments on social media are collected for analysis. What type of data is this?
14. A company collects a very large amount of daily user activity data from its mobile app. Which Big Data characteristic is applied?
15. A hospital dataset contains missing age values and incorrect phone numbers. Which data quality issue is present?
16. A business wants to convert customer data into useful sales decisions. Which Big Data characteristic is important?
17. A company stores large files in Hadoop Distributed File System. Which part of the Big Data ecosystem is used?
18. A company wants to process large datasets using parallel computation. Which Big Data tool can be used?
19. An organization wants to collect live sensor data from IoT devices. Which Big Data ecosystem component is suitable?
20. A company wants to visualize Big Data results for decision-makers. Which tool or method can be applied?

① Predict Student failure

Data science analyzes students attendance, marks and assignments to predict which students may fail. Teachers can then provide extra support to improve their performance.

② Analyze Hospital Records

Data science analyze patient records to predict diseases, improve treatment plans and help doctors make better medical decisions.

③ Recommend Products.

This is a Recommendation system. It suggests products & based on customer previous purchase & browsing history.

④ Detect Bank fraud.

Data science analyzes transaction patterns & identifies unusual activities to detect fraud & prevent financial losses.

⑤ Reduce customer loss.

Data science predicts which customer are likely to leave & helps the company take actions such as offering discounts or better services.

⑥ Improve Student Performance.

Data science helps identify weak students early so teacher can provide additional guidance & improve academic performance.

⑦ Festival Sales analysis.

Data science analyze past sales data to identify popular products during festivals & helps the store manage inventory effectively.

⑧ Reduce traffic congestion

Data Science analyzes traffic data to identify busy routes & improve traffic signal timing, reduce congestion.

⑨ Improve crop yield

Data Science analyzes weather, soil, & crop data to recommend best time for planting, watering & harvesting.

⑩ Different data formats

This shows variety, because the data is available in different formats such as Structured, Semi-Structured & Unstructured data.

⑪ Fast data updates

This represents velocity, as data is generated & updated continuously at high speed.

⑫ Rows & columns data

This is structured data, because the information is organized in tables with rows and columns.

⑬ Social media comments

This is unstructured data because comments are free from text without a fixed format.

⑭ Large amount of data

This represents volume, because a huge amount of data is generated & stored every day.

⑮ Useful Business insights, Missing & Incorrect data

This is Data Quality issue because data set contains incomplete & incorrect information.

⑩ Useful Business Insights

The important characteristic is value as goal is to extract useful insights from data for better business decisions.

⑪ Store large files.

company is using HDFS which stores large amounts of data across multiple computers.

⑫ Parallel data processing

Apache spark can be used because it processes large datasets quickly using Parallel computing.

⑬ collect live IoT data

Apache Kafka is suitable because it collects & manages real time streaming data from IoT devices

⑭ visualize Big data.

Tools such as Tableau (or) Power BI can be used to create charts, dashboards and reports for easy decision making.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Criteria	Weightage	No. of Questions	Marks Obtained
Conceptual Understanding	30	6	
Accuracy of Answer	25	5	
Application of Knowledge	20	4	
Completeness of Response	15	3	
Clarity & Presentation	10	2	
TOTAL			